Formal Logic (SM234614)

First Meeting: Boolean Logic

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- Conjunction (AND, $\wedge$)

$$\wedge : (\mathbb{B}, \mathbb{B}) \to \mathbb{B}$$

$a \wedge b$ yields 1 if both a and b are 1; 0 otherwise

- Negations (NOT, $\neg$)

$$\neg : \mathbb{B} \to \mathbb{B}$$

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$$a \vee b := \neg(\neg a \wedge \neg b)$$

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$$a \vee b := \neg(\neg a \wedge \neg b)$$

$a \wedge b$ yields 0 if both a and b are 0; 1 otherwise

- Implication (IF, $\rightarrow$)

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$$a \leftrightarrow b := \neg(\neg(a \wedge b) \wedge \neg(\neg(a) \wedge \neg(b)))$$

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- Etc (NOR, XOR, ...)

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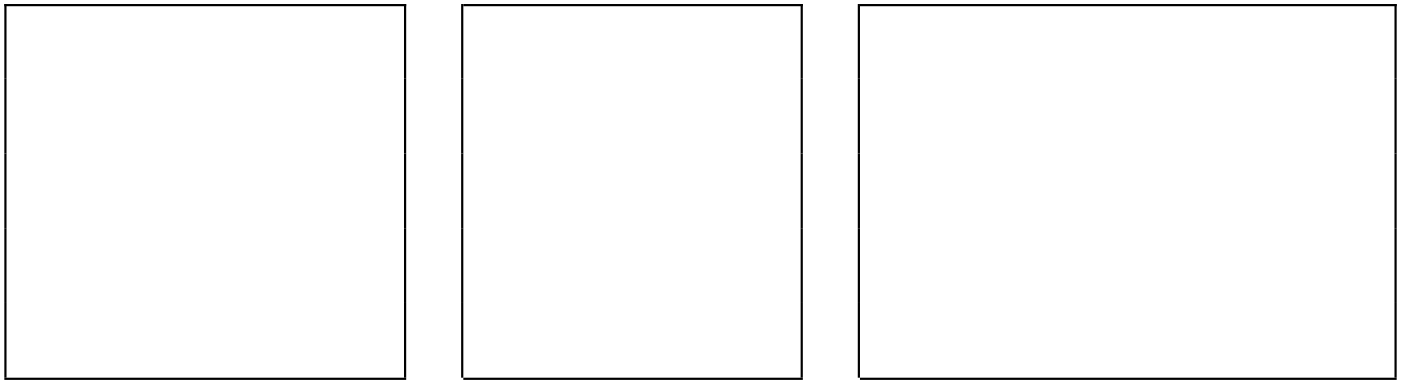
$$a \to b \equiv \neg(a \wedge \neg b) \equiv \neg a \vee b \equiv \neg a + b$$

$$a \leftrightarrow b \equiv \neg(\neg(a \wedge b) \wedge \neg(\neg(a) \wedge \neg(b))) \equiv ab + \neg a \neg b$$

Several properties



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$$\neg\neg a = a$$
$$\neg(a + b) = \neg a \cdot \neg b$$
$$\neg(a \cdot b) = \neg a + \neg b$$

Distributive law

$$a \cdot (b + c) = a \cdot b + a \cdot c$$
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Given a set $\mathbb{B} = \{0, 1\}$. The syntax of Boolean function is defined as follows

$$f :: x \mid \neg(f) \mid f_1 \wedge f_2 \mid f_1 \vee f_2$$

where x is a variable in $\mathbb{B}$.

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Example:

- $f(x_1, x_2, x_3) = x_1x_2 + \neg x_1x_3 + x_2\neg x_3$
- $g(x_1, x_2, x_3) = x_1 + x_2 + x_2x_3$
- $h(x_1, x_2, x_3) = 1$

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The function $g(x_1, x_2) = (x_1x_2 + \neg x_1\neg x_2)(x_1\neg x_2 + \neg x_1x_2)$ is non-satisfiable

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Models for f: $(0, 1), (1, 0), (1, 1)$
Models for g: $(0, 1, 1), (1, 0, 0), (1, 1, 0), (1, 1, 1)$

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- A Boolean function is in Disjunctive Normal Form (DNF) if it is a disjunction of several terms: $ab, ab + c\neg d$
- A Boolean function in CNF form can be converted into DNF form; and vice versa.

$$
\begin{aligned}
(a+b)(a+c+d)(b+\neg d) &\equiv (a+ac+ad+ab+bc+bd)(b+\neg d) \\
&\equiv (a+bc+bd)(b+\neg d) \\
&\equiv ab+a\neg d+bc+bc\neg d+bd+0 \\
&\equiv ab+a\neg d+bc+bd
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$$

$$
\begin{aligned}
a_1b_1+a_2b_2+\ldots+a_nb_n \equiv &(a_1+a_2+\ldots+a_n)(b_1+a_2+\ldots+a_n) \\
&(a_1+b_2+\ldots+a_n)\ldots \\
&\vdots \\
&(b_1+b_2+\ldots+a_n)(b_1+b_2+\ldots+b_n)
\end{aligned}
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&\vdots \\
&(b_1+b_2+\ldots+a_n)(b_1+b_2+\ldots+b_n)
\end{aligned}
$$

In general, converting CNF to DNF is easier than converting DNF to CNF.

p	q	r	$(p \vee \neg q) \rightarrow r$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
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$$(p \vee \neg q) \rightarrow r \equiv \neg p \neg q r + \neg p q \neg r + \neg p q r + p \neg q r + p q r$$

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$$(p \vee \neg q) \rightarrow r \equiv \neg p \neg q r + \neg p q \neg r + \neg p q r + p \neg q r + p q r$$
$$\equiv (p + q + r)(\neg p + q + r)(\neg p + \neg q + r)$$

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  - Define new clauses for each variable
  - Apply the following rules

$$p \leftrightarrow (a \vee b) \equiv (p \vee \neg a) \wedge (p \vee \neg b) \wedge (\neg p \vee a \vee b)$$
$$p \leftrightarrow (a \wedge b) \equiv (\neg p \vee a) \wedge (\neg p \vee b) \wedge (p \vee \neg a \vee \neg b)$$
$$p \leftrightarrow \neg a \equiv (p \vee a) \wedge (\neg p \vee \neg a)$$

$$f = ((p \vee q) \wedge r) \vee \neg p$$

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$$x_3 := \neg p$$

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$$CNF(f) := x_4 \wedge (x_4 \leftrightarrow (x_2 \vee x_3)) \wedge (x_3 \leftrightarrow \neg p) \wedge (x_2 \leftrightarrow (x_1 \wedge r)) \wedge (x_1 \leftrightarrow (p \vee q))$$

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$$x_1 := p \vee q$$
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$$\begin{aligned} CNF(f) :=& x_4 \wedge (x_4 \leftrightarrow (x_2 \vee x_3)) \wedge (x_3 \leftrightarrow \neg p) \wedge \\ & (x_2 \leftrightarrow (x_1 \wedge r)) \wedge (x_1 \leftrightarrow (p \vee q)) \\ :=& x_4 \wedge (x_4 \vee \neg x_2) \wedge (x_4 \vee \neg x_3) \wedge (\neg x_4 \vee x_2 \vee x_3) \wedge \\ & (x_3 \vee p) \wedge (\neg x_3 \vee \neg p) \wedge \\ & (\neg x_2 \vee x_1) \wedge (\neg x_2 \vee r) \wedge (x_2 \vee \neg x_1 \vee \neg r) \wedge \\ & (x_1 \vee \neg p) \wedge (x_1 \vee \neg q) \wedge (\neg x_1 \vee p \vee q) \end{aligned}$$